Auditoría, Calidad y Fiabilidad Informáticas

(IT Auditing, Quality, and Reliability)

Universidad Complutense de Madrid Academic Year 2026/27

End-of-Term Survey on the Maude Project

English translation of the instrument administered in Spanish. Item numbers match the Spanish version.

Before you start. This survey is **anonymous** and **voluntary**: it carries no marks, there is no record of who answered, and it does not affect anyone's grade. It takes about 10 minutes. This year the course was assessed differently from previous years, and your answers here are the only way to tell whether the change was worth it or should be undone. Please take your time and do not sugar-coat anything: a critical answer is more useful than a kind one.

Use of the data. Your answers may be published in a paper about the redesign of this course, including individual answers and verbatim quotes from the open questions. Your name, and any data that would identify you, will not be published. If you prefer your answers not to be used for that purpose, tick the box in the last block.

A. Context

1. How many people were in your team?
   ❑ Two ❑ Three

2. The system you modeled was:
   ❑ From the list in the assignment ❑ Proposed by us

3. How much did you know about that system *before* starting the project?
   ❑ Nothing ❑ I had heard of it ❑ I had used or studied it superficially ❑ I knew it well

4. *(Optional)* Which system did you model? _____
   You may leave this blank: if yours was the only team that chose that system, answering here makes your team identifiable (though not you within it).

B. Time spent

5. Approximately how many hours **in total** did you spend on the project outside class?
   ❑ Fewer than 10 ❑ 10–19 ❑ 20–29 ❑ 30–39 ❑ 40 or more

6. Roughly how was that time distributed? (give an approximate percentage per row; it should add up to 100)

Understanding the real system (documentation, papers, code)	_____ %
Writing and debugging the Maude specification	_____ %
Stating and checking the properties (`search`, LTL)	_____ %
Mutations	_____ %
Writing the report and preparing the defense	_____ %

7. Compared with what you expected when you read the assignment, the actual workload was:
   ❑ Much lower ❑ Lower ❑ As expected ❑ Higher ❑ Much higher

C. Perceived learning

Indicate how much you agree with each statement: 1 = strongly disagree, 2 = disagree, 3 = neither agree nor disagree, 4 = agree, 5 = strongly agree.

	1	2	3	4	5
8. The project made me connect ideas from this course with knowledge from other courses.	❑	❑	❑	❑	❑
9. I had to analyze an idea or a mechanism in depth, not just apply a recipe.	❑	❑	❑	❑	❑
10. I learned something that changed my understanding of how real concurrent systems work.	❑	❑	❑	❑	❑
11. I am now better at deciding when something should be an equation and when a rule.	❑	❑	❑	❑	❑
12. I now understand better what a formal verification does and does *not* guarantee.	❑	❑	❑	❑	❑
13. I would be able to apply what I learned to a system other than the one my team modeled.	❑	❑	❑	❑	❑
14. Deciding how to represent the system (which sorts and constructors to use, what is state and what is a message) was harder for me than writing the equations and rules.	❑	❑	❑	❑	❑

D. Engagement with the project

For each statement, indicate how well it describes you: 1 = does not describe me at all, 2 = describes me a little, 3 = describes me quite well, 4 = describes me totally.

	1	2	3	4
15. In the sessions devoted to the project I worked with concentration.	❑	❑	❑	❑
16. I put more effort into the project than the minimum needed to pass it.	❑	❑	❑	❑
17. I enjoyed working on the project.	❑	❑	❑	❑
18. When something did not work, I wanted to keep at it until I understood why.	❑	❑	❑	❑
19. I found the project boring.	❑	❑	❑	❑
20. I did just enough to meet the requirements of the assignment.	❑	❑	❑	❑
21. I have talked about the project with people outside the course.	❑	❑	❑	❑
22. I was left wanting to keep improving the model after the submission.	❑	❑	❑	❑

E. Comparison with the short practicals earlier in the course

Before the project you did three short Maude practicals in this same course, which did not count towards the grade. Compared with those practicals, I found the project:
1 = much less, 2 = somewhat less, 3 = the same, 4 = somewhat more, 5 = much more.

	1	2	3	4	5
23. ... interesting	❑	❑	❑	❑	❑
24. ... demanding	❑	❑	❑	❑	❑
25. ... useful for actually learning Maude	❑	❑	❑	❑	❑
26. ... related to what I expect to do professionally	❑	❑	❑	❑	❑
27. ... demanding in terms of understanding the system *before* being able to write any code	❑	❑	❑	❑	❑

28. If you could choose the assessment format for this course, you would prefer:
❑ Only short practicals like the ones early in the course ❑ Only a team project like this one
❑ Both, as this year ❑ No preference

F. AI assistants and the MCP for Maude

Using AI assistants on this project was **permitted**, and the MCP for Maude recommended, so this block is not about catching anything: what matters is how they worked for you and, above all, whether the MCP for Maude adds something a generic assistant does not. This block is anonymous, like the rest.

29. How often did you use an assistant during the project?
❑ Never ❑ Occasionally ❑ Frequently ❑ Almost constantly

30. Which ones? *(tick all that apply)*
❑ ChatGPT ❑ Claude ❑ Copilot ❑ Gemini ❑ Other: _____

31. What did you use them for? *(tick all that apply)*

❑ Understanding the real system we were going to model
❑ Looking up Maude syntax
❑ Explaining error messages from the interpreter
❑ Writing specific equations or rules
❑ Writing the whole specification, or almost all of it
❑ Stating the properties (`search`, LTL)
❑ Interpreting the model checker's output
❑ Proposing the mutations
❑ Writing parts of the report
❑ Other: _____

32. Did you use the MCP for Maude (Harold)?
❑ I did not install it ❑ I installed it but hardly used it ❑ I used it sometimes ❑ I used it regularly

If you did not use it, briefly say why and skip to question 38: _____

33. Installing it and getting it running was easy:
❑ 1 (not at all) ❑ 2 ❑ 3 ❑ 4 ❑ 5 (a lot)

34. The diagnostics it returns about the specification were useful to me:
❑ 1 (not at all) ❑ 2 ❑ 3 ❑ 4 ❑ 5 (a lot)

35. With the MCP, the assistant generated more correct Maude code than without it:
❑ 1 (not at all) ❑ 2 ❑ 3 ❑ 4 ❑ 5 (a lot) ❑ I cannot compare

36. In which situations did you find the MCP most useful?

_____

37. What is missing from the MCP, or what would you change?

_____

_____

38. Of the Maude code an assistant generated for you, how much did you have to redo or fix?
❑ I did not use assistants for code ❑ Nothing or almost nothing ❑ A little ❑ Quite a lot ❑ Almost all of it

39. Did you understand all the code you submitted, wherever it came from?
❑ Yes, all of it ❑ Almost all ❑ There were parts I did not quite understand ❑ There were quite a few parts I did not understand

40. Did you try asking an assistant for the whole project from the assignment text, just to see what came out?
❑ We did not try ❑ Yes, and it produced nothing usable ❑ Yes, and it produced something that had to be almost entirely redone ❑ Yes, and it worked reasonably well

41. Compared with the short practicals earlier in the course, solving this project with the help of an AI seemed:
❑ Much easier ❑ Easier ❑ The same ❑ Harder ❑ Much harder

42. What do the assistants do worst in Maude?

G. To finish

43. What is the best thing about the project? What would you keep?

44. What is the worst? What would you change?

45. What advice would you give someone starting this project next year?

46. ❑ I do **not** want my answers to be used in a publication about the design of this course.

Thank you for your time.